

A Dimension-Four Twisted-Convolution Identity from a Shintani Ray Unit

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Abstract

We study the twisted-convolution condition that turns canonical Shintani–Faddeev real-multiplication values into ghost projectors for Weyl–Heisenberg symmetric informationally complete measurements. We give exact Zak-rank, exterior-square, and parity-moment formulations for the canonical family. In dimension four, all 36 shifted ghost minors have the common factor $x^2 - \sqrt{3 + \sqrt{5}}x + 1$. Using the Shintani–Kopp Kronecker-limit formula and the analytic class-number formula for $\mathbb{Q}(\sqrt{5}, \sqrt{\phi})$, we identify $x^2 = \phi + \sqrt{\phi}$, where $\phi = (1 + \sqrt{5})/2$. Thus the canonical dimension-four ghost has rank one and satisfies twisted convolution. General-dimensional twisted convolution remains open.

1 Canonical setup

For $d \geq 4$, put

$$Q_d = \langle 1, 1 - d, 1 \rangle, \quad L_d = \begin{pmatrix} d-1 & -1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$\text{disc}(Q_d) = (d+1)(d-3), \quad L_d^2 + L_d + I = dL_d,$$

so $L_d^3 \equiv I \pmod{d}$. Its positive quadratic fixed point is

$$\beta_d = \frac{d-1 + \sqrt{(d+1)(d-3)}}{2}.$$

For $D_{p,q} = \tau^{pq} X^p Z^q$, $\tau = -e^{\pi i/d}$, write

$$P = \frac{1}{d} \sum_{\mathbf{p}} a_{\mathbf{p}} D_{\mathbf{p}}.$$

The equation $P^2 = P$ is the finite twisted-convolution equation for the coefficients. In even dimensions, the representative-wrap signs of the displacement operators are retained.

2 Finite rank certificates

After a scalar shift, shear, and partial Fourier transform, canonical twisted convolution is equivalent to rank one of a $d \times d$ matrix K_d . Thus it is equivalent to the vanishing of all 2-minors. For any complex matrix K ,

$$\Delta_2(K) = \frac{1}{2} \left((\text{Tr } K^\dagger K)^2 - \text{Tr}((K^\dagger K)^2) \right) = \sum_{\substack{|I|=2 \\ |J|=2}} |\det K_{I,J}|^2.$$

Hence $\Delta_2(K) = 0$ if and only if $\text{rank } K \leq 1$.

The ghost also obeys parity-Hermiticity $G^\dagger = PGP$. Therefore $J = PG$ is Hermitian and the rank-one certificate becomes

$$\text{Tr } J^4 = (\text{Tr } J^2)^2.$$

Reciprocity forces lower trace information but does not force this fourth moment.

3 Dimension four

Let

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad \beta = \phi^2, \quad \tau = -e^{\pi i/4}.$$

The normalized signed double-sine table is

$$\begin{pmatrix} \sqrt{5} & -x & 1 & -x^{-1} \\ -x^{-1} & -x^{-1} & -x^{-1} & -x \\ 1 & -x^{-1} & 1 & x \\ -x & -x^{-1} & x & -x \end{pmatrix}.$$

The zero characteristic is exceptional and is replaced by $\sqrt{5}$ before applying the displacement chirp.

Proposition 1. *Every one of the 36 two-by-two minors of the shifted dimension-four ghost matrix lies in the principal ideal*

$$\left(x^2 - \sqrt{3 + \sqrt{5}}x + 1 \right)$$

of $\mathbb{Q}(\sqrt{2}, \sqrt{5})[x, x^{-1}]$.

Proof. The exact Laurent expansion has 34 formally nonzero minors and two structurally zero minors. Reduce each power using

$$x^{-1} = t - x, \quad x^2 = tx - 1, \quad t = \sqrt{3 + \sqrt{5}} = \frac{\sqrt{2} + \sqrt{10}}{2}.$$

Every remainder $A + Bx$ is zero. The deterministic certificate is generated by `scripts/generate_referee_certificates.py`. $\square$

It remains to prove $x + x^{-1} = \sqrt{3 + \sqrt{5}}$. Shift and reflection give

$$x = \sqrt{2} \frac{S_2(\beta/4 \mid \beta, 1) S_2(1/4 \mid \beta, 1)}{S_2((\beta + 1)/4 \mid \beta, 1)}.$$

4 The modulus-four ray field

Let $K = \mathbb{Q}(\sqrt{5})$ and $\mathcal{O}_K = \mathbb{Z}[\phi]$. Since 2 is inert, finite residue enumeration gives

$$|\text{Cl}_{(4)\infty_2}(K)| = 2.$$

Set $L = K(\sqrt{\phi})$. Its relative discriminant is (4), and the second real place ramifies because $\phi' < 0$. Hence L is the full ray field of modulus $(4)\infty_2$.

Put $u = \phi + \sqrt{\phi}$. Its Artin conjugate is $u^\sigma = \phi - \sqrt{\phi} = u^{-1}$, so

$$N_{L/K}(u) = 1, \quad \text{Tr}_{L/K}(u) = 1 + \sqrt{5}.$$

5 Class number, units, and regulator

The field L has signature $(2, 1)$ and absolute discriminant -400 . Its Minkowski bound is $15/(2\pi) < 2.4$. There is no ideal of norm two, hence $h_L = 1$.

Write $t = \sqrt{\phi}$, so $t^4 - t^2 - 1 = 0$. The units t and $u = t^2 + t$ form a fundamental system. A centered logarithmic parallelogram argument bounds a hypothetical missing unit at all archimedean places. Exact norm enumeration in the integral basis $1, t, t^2, t^3$ leaves only ± 1 in that cell. Consequently

$$\mathcal{O}_L^\times = \{\pm t^m u^n : m, n \in \mathbb{Z}\}, \quad R_L = \log \phi \log u.$$

PARI/GP independently returns discriminant -400 , class number one, ray structures $[2]$ and $[2, 2]$, and the same regulator; its transcript is included in `certificates/pari-audit.txt`.

For the nontrivial quadratic character χ of L/K ,

$$L(s, \chi) = \frac{\zeta_L(s)}{\zeta_K(s)}.$$

Since $h_K = h_L = 1$, $w_K = w_L = 2$, and $R_K = \log \phi$, the analytic class-number formula at zero yields

$$L'(0, \chi) = \log u.$$

6 Kronecker-limit normalization

The two-infinite-place ray group has structure $[2, 2]$, while the one-infinite-place group has structure $[2]$. The quotient fibers therefore have order two, so the exponent in Kopp's theorem is $n = 1$. For the two ray classes A_0, A_1 ,

$$L(s, \chi) = \zeta(s, A_0) - \zeta(s, A_1).$$

The generalized derivative in the Shintani–Kopp formula is this difference, and the normalized positive cocycle occurs squared. Therefore

$$x^2 = \exp L'(0, \chi) = u = \phi + \sqrt{\phi}.$$

Theorem 2. *Assuming the published principal-ghost formula and the proved Shintani–Kopp Kronecker-limit theorem with their stated conventions, the canonical dimension-four ghost has rank one and satisfies twisted convolution. No Stark conjecture is assumed.*

Proof. Because $x^2 = u$ and $u + u^{-1} = 1 + \sqrt{5}$,

$$(x + x^{-1})^2 = 3 + \sqrt{5}.$$

The defining double-sine value is positive, hence $x + x^{-1} = \sqrt{3 + \sqrt{5}}$. The common minor factor vanishes, so all 36 minors vanish. The shifted Zak matrix has rank one, which is equivalent to normalized ghost idempotency and twisted convolution. $\square$

7 Scope and review points

The theorem concerns the canonical dimension-four ghost. It does not prove general TCC, the Minimalist RM Values Conjecture, the Stark conjecture, or Zauner's conjecture in arbitrary dimension.

Before submission, a referee should independently audit: (i) the reciprocal double-sine convention; (ii) the exceptional zero characteristic; (iii) the exact 36-minor certificate; (iv) the ray class and infinite-place convention; (v) the cocycle-square normalization in Kopp's theorem; and (vi) the final projector normalization.

References

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- [2] T. Shintani, *On a Kronecker limit formula for real quadratic fields*, J. Fac. Sci. Univ. Tokyo 24 (1977), 167–199.
- [3] T. Shintani, *On certain ray class invariants of real quadratic fields*, J. Math. Soc. Japan 30 (1978), 139–167.
- [4] S. Flammia et al., *Zauner.jl*, <https://github.com/sflammia/Zauner.jl>.